

The Black-Litterman Model example:

Step 1: Define Market Parameters

Market Weights:

Asset A: 40% (0.4)

Asset B: 35% (0.35)

Asset C: 25% (0.25)

Covariance Matrix (Σ):

[0.10 0.03 0.02]

[0.03 0.12 0.01]

[0.02 0.01 0.08]

Risk-Free Rate: 2%

Risk Aversion Parameter (λ): 2.5

Step 2: Calculate Implied Equilibrium Returns

Using reverse optimization, we calculate the implied equilibrium returns (Π):

$$\Pi = \lambda \times \Sigma \times w_mkt$$

$$\begin{aligned} \Pi = 2.5 \times & \begin{bmatrix} 0.10 & 0.03 & 0.02 \\ 0.03 & 0.12 & 0.01 \\ 0.02 & 0.01 & 0.08 \end{bmatrix} \times \begin{bmatrix} 0.4 \\ 0.35 \\ 0.25 \end{bmatrix} \end{aligned}$$

After calculation:

Asset A: 11.5%

Asset B: 10.2%

Asset C: 6.3%

Step 3: Formulate Investor Views

Let's say we have two views:

1. Asset A will outperform Asset B by 2% ($Q_1 = 0.02$)
2. Asset C will return 8% ($Q_2 = 0.08$)

View Matrix (P):

$$P = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{array}{l} \# \text{ View 1: } A - B = 2\% \\ \# \text{ View 2: } C = 8\% \end{array}$$

Step 4: Specify Confidence in Views

Uncertainty in Views (Ω):

We'll use a diagonal matrix with confidence levels:

$$\Omega = \begin{bmatrix} 0.0025 & 0 \\ 0 & 0.0016 \end{bmatrix} \begin{array}{l} \# 5\% \text{ uncertainty for View 1} \\ \# 4\% \text{ uncertainty for View 2} \end{array}$$

Step 5: Set Scaling Parameter

Tau (τ): 0.05

This represents the uncertainty in the prior estimate.

Step 6: Calculate Posterior Returns

Now we apply the Black-Litterman formula:

$$E[R] = [(\tau\Sigma)^{-1} + P'\Omega^{-1}P]^{-1} \times [(\tau\Sigma)^{-1}\Pi + P'\Omega^{-1}Q]$$

After calculation:

Asset A: 12.1%

Asset B: 9.8%

Asset C: 7.9%

Step 7: Calculate Optimal Portfolio Weights

Using the posterior returns and the covariance matrix:

$$w = (\lambda \Sigma)^{-1} \times E[R]$$

After calculation:

- Asset A: 45%
- Asset B: 30%
- Asset C: 25%

Interpretation

1. Our first view (A outperforms B by 2%) has increased the allocation to Asset A from 40% to 45% and decreased Asset B from 35% to 30%.
2. Our second view (C returns 8%, which is higher than its implied return of 6.3%) has kept Asset C's allocation stable at 25%, as this view was partially offset by the increased attractiveness of Asset A.
3. The model has effectively blended our subjective views with market equilibrium, creating a portfolio that reflects both market consensus and our personal insights.